

KAMAL ACHARYA

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RESEARCH SUMMARY

AI researcher specializing in trustworthy and neurosymbolic artificial intelligence, with emphasis on integrating machine learning, symbolic knowledge, and reasoning to develop interpretable, robust, and knowledge-guided decision-support systems. Research contributions span symbolic knowledge distillation of large language models, neurosymbolic reinforcement learning, explainable predictive modeling, and data-driven demand forecasting using multimodal and spatiotemporal data. Published in leading venues including IEEE Transactions on Artificial Intelligence, IEEE Transactions on Intelligent Transportation Systems, IEEE Communications Surveys & Tutorials, IJCAI, and AIAA. Experienced in NASA and NSF funded multidisciplinary research, student mentorship, and university-level teaching.

RESEARCH HIGHLIGHTS

- 17+ peer-reviewed publications, including first-author papers in IEEE Transactions on Artificial Intelligence, IEEE Transactions on Intelligent Transportation Systems, IEEE Communications Surveys & Tutorials, IJCAI, IEEE CAI, IEEE ITSC, and AIAA.
- Research conducted within NASA University Leadership Initiative (ULI) and National Science Foundation (NSF) funded projects.
- Research contributions in trustworthy AI, neurosymbolic AI, symbolic knowledge distillation, explainable machine learning, and transportation demand modeling.
- Reviewer for leading journals including IEEE Transactions on Artificial Intelligence, IEEE Transactions on Intelligent Transportation Systems, IEEE Transactions on Neural Networks and Learning Systems, IEEE Internet of Things Journal, and Journal of Big Data.
- IEEE Senior Member, IEEE Baltimore Section Vice-Chair Elect, and Co-Chair of the IEEE Baltimore Technical Colloquium and Professional Development Conference 2026.

RESEARCH INTERESTS

Core Research Areas: Trustworthy and Explainable AI; Neurosymbolic AI; Large Language Models and Foundation Models; Symbolic Knowledge Distillation; Neurosymbolic Reinforcement Learning; Knowledge Representation and Reasoning; Multimodal and Spatiotemporal Machine Learning.

Application Areas: Intelligent Decision Support Systems; Intelligent Transportation Systems; Advanced Air Mobility; Transportation Demand Modeling and Forecasting; AI for Complex Cyber-Physical Systems.

TECHNICAL SKILLS

Programming	Python, C/C++, SQL, PHP
AI Frameworks	PyTorch, TensorFlow, Scikit-learn
Data Science	NumPy, Pandas, GeoPandas, MATLAB, Jupyter Notebook
Development Tools	Linux, Docker, Git, CUDA, MySQL
Research Expertise	Trustworthy AI, Explainable AI, Neurosymbolic AI, Large Language Models, Reinforcement Learning, Symbolic Knowledge Distillation, Predictive Modeling, Spatiotemporal Modeling, Transportation Demand Modeling, Statistical Learning
Languages	English (Fluent), Nepali (Native), Hindi (Fluent)

EDUCATION

Ph.D., Information Systems

May 2026

University of Maryland, Baltimore County (UMBC)

Dissertation: Enhancing Demand Modeling for Advanced Air Mobility Using Data-Driven Learning and Neurosymbolic AI

Advisor: Dr. Houbing H. Song

Co-Advisor: Dr. Alvaro Velasquez

Dissertation Committee: Dr. Nirmalya Roy, Dr. Lei Zhang, Dr. Liang Sun, Dr. Kai Sun

M.Sc., Information System Engineering

2019

Himalayan Institute of Science and Technology (HIST), Purbanchal University

Thesis: An Analysis of Classification Algorithms for Nepali News

Supervisor: Dr. Subarna Shakya

B.E., Electronics and Communication Engineering

2011

Himalaya College of Engineering (HCOE), Tribhuvan University

Final Year Project: Distance and Speed Measurement Using Ultrasound

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate

Aug 2026 – Present

Baylor University, Waco, TX, USA

- Conduct research on trustworthy artificial intelligence, machine learning, and data-driven decision support systems.
- Develop AI models and computational methodologies for predictive modeling, optimization, and intelligent decision-making.
- Design research experiments, analyze large-scale datasets, and validate model performance using statistical and machine learning techniques.
- Prepare manuscripts for leading journals and conferences and contribute to competitive external research grant proposals.
- Mentor graduate and undergraduate researchers while supporting collaborative interdisciplinary research initiatives.

Graduate Research Assistant

Aug 2023 – May 2026

University of Maryland, Baltimore County, Baltimore, MD, USA

- Conducted doctoral research under the NASA University Leadership Initiative (ULI) on mobility–energy coordinated infrastructure planning for Advanced Air Mobility (AAM).
- Developed scalable machine learning and neurosymbolic AI methods for transportation demand forecasting, trip generation, and regional air mobility planning using large-scale spatiotemporal datasets.
- Designed a rolling-horizon trip request generation framework for time-dependent AAM demand simulation, supporting infrastructure planning and operational decision-making.
- Developed interpretable predictive models integrating machine learning and symbolic reasoning for transportation demand estimation.
- Published multiple first-author papers in IEEE Transactions on Intelligent Transportation Systems, IJCAI, IEEE CAI, IEEE ITSC, IEEE Big Data, IWCMC, and AIAA based on this research.

Teaching Assistant

Operationalizing AI/Machine Learning for Cybersecurity Training

May 2024 – Aug 2024

University of Maryland, Baltimore County (MD, USA)

- Supported a multidisciplinary graduate training program on Artificial Intelligence, Machine Learning, and Cybersecurity supported by NSF grants (OAC-2309760 and OAC-2229976).
- Assisted instructors in delivering hands-on instruction on AI-driven cybersecurity and cyberinfrastructure tools.
- Mentored interdisciplinary student teams conducting research projects integrating AI and cybersecurity applications.

- Provided guidance on project implementation, technical reports, and final research presentations.

Research Assistant

Aug 2022 – Aug 2023

Embry-Riddle Aeronautical University (FL, USA)

- Worked on the Center for Advanced Transportation Mobility (CATM) project: “Improving Air Mobility in Emergency Situations”.
- Data pre-processing and visualization: data gathering, cleansing, integration, and visualization.
- Developed deep learning models for micro-level and macro-level air mobility forecasting and transportation system analysis.
- Developed learning-to-optimize frameworks for air mobility operations and emergency transportation planning.

Information Technology Coordinator

2014 – 2021

College of Applied Business and Technology (Kathmandu, Nepal)

- Helped establish a research and development unit in the IT sector in the college.
- Taught multiple undergraduate computing and programming courses within the institution.
- Conducted guest lectures and special seminars; supervised and coordinated faculty duty rosters.
- Counseled students; hired IT-related faculty in consultation with leadership.
- Developed and administered non-credit IT courses; organized annual IT symposium/workshops and competitions.
- Encouraged student publications (magazines, newspapers, journals).

Web Developer

2013 – 2014

Uniweb Technologies (Kathmandu, Nepal)

- Developed and maintained websites/web applications using HTML, CSS, JavaScript, and PHP.
- Designed responsive interfaces for cross-device/browser compatibility.
- Managed and optimized databases using complex SQL queries; improved performance via refactoring and caching.
- Integrated third-party APIs (payment gateways, social media); performed unit/integration/UAT testing.

Visiting Lecturer

2012 – 2021

Various Colleges Affiliated to Tribhuvan and Purbanchal Universities (Kathmandu, Nepal)

- Taught: C, C++, Computer Organization, Computer Architecture, Digital Logic, Advanced Internetworking.
- Supervised major and minor projects; involved in BIM, BCA, and BIT programs.

Lecturer

2012 – 2013

Janakpur Engineering College (Kathmandu, Nepal)

- Delivered assigned courses; encouraged assignments and student counseling.
- Guided field work; engaged students in research/projects; prepared detailed course plans.
- Supervised minor and major projects.

RESEARCH FUNDING

- NASA University Leadership Initiative (ULI): Mobility–Energy Coordinated Platform for Advanced Air Mobility Operations — Doctoral Research Contributor.
- National Science Foundation (NSF): AI/ML Cybersecurity Training Program (Grants OAC-2309760, OAC-2229976) — Teaching Assistant.
- Experience working within large-scale, multi-institutional funded research collaborations.

SELECTED RESEARCH PROJECTS

- **NASA ULI: Mobility–Energy Coordinated Platform for Advanced Air Mobility Operations**
Developed scalable demand forecasting and trip generation frameworks for regional air mobility systems, supporting infrastructure planning and transportation-energy coordination.

- **Demand Modeling for Advanced Air Mobility**

Designed data-driven and theory-guided approaches for multimodal transportation demand estimation using generalized cost modeling and machine learning methods.

- **Improving Air Mobility in Emergency Situations**

Developed AI-driven forecasting and optimization models for emergency and disaster-response air mobility planning under the CATM initiative.

PUBLICATIONS

Journal

- Vasiloff K, **Acharya K**, Wang Z, Song H, Sun L, "Demand Forecast and Energy-Aware Portal Siting for Regional Air Mobility" (**Under Major Revision**), *Journal of Air Transport Management*, 2026
- **Acharya K**, Raza W, Vasiloff K, Wang Z, Sun L and Song H, "Demand Modeling for Advanced Air Mobility: Challenges, Opportunities, and Future Directions", in *IEEE Transactions on Intelligent Transportation Systems*, doi: 10.1109/TITS.2026.3671002.
- **Acharya K**, Song H, "A Comprehensive Review of Neuro-symbolic AI for Robustness, Uncertainty Quantification, and Intervenability", *Arab J Sci Eng* (2025). <https://doi.org/10.1007/s13369-025-10887-3> (**Invited Paper**)
- De Macêdo AR, Jagatheesaperumal SK, Da Costa KA, **Acharya K**, Song H, Guizani M, De Albuquerque VH, "Quantum AI-Enhanced IoT-Fog Communication: A Survey From Cybersecurity and Data Privacy Perspective", in *IEEE Communications Surveys & Tutorials*, vol. 28, pp. 3637-3665, 2026, doi: 10.1109/COMST.2025.3622378.
- **Acharya K**, Velasquez A and Song H, "A Survey on Symbolic Knowledge Distillation of Large Language Models", in *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 12, pp. 5928-5948, Dec. 2024, doi: 10.1109/TAI.2024.3428519
- **Acharya K**, Raza A, Dourado C, Velasquez A and Song H, "Neurosymbolic Reinforcement Learning and Planning: A Survey", in *IEEE Transactions on Artificial Intelligence*, vol. 5, no. 5, pp. 1939-1953, May 2024, doi: 10.1109/TAI.2023.3311428. (**Selected for IEEE CIS Publication Spotlight, Nov. 2024**)
- Raza W, Ma X, Song H, Ali A, Zubairi H, **Acharya K**, "Long Short-Term Memory Neural Network assisted Peak to Average Power Ratio Reduction for Underwater Acoustic Orthogonal Frequency Division Multiplexing Communication", *KSII Transactions on Internet & Information Systems*. 2023 Jan 1;17(1), doi: 10.3837/tiis.2023.01.013

Book Chapter

- Hakim S.B, Adil M, **Acharya K** and Song H, "AI for Android Malware Detection and Classification" , in *AI for Cybersecurity* (eds H.H. Song, E. Bertino, A. Velasquez, H.H. Wang, Y. Shoshitaishvili and S.K. Jha). <https://doi.org/10.1002/9781394293773.ch15>

Conference Proceedings

- **Acharya K**, Vasiloff K, Wang Z, Song H, Sun L, "Urban Air Mobility Flight Demand Modeling for Airports in New York City", In *AIAA SCITECH 2026 Forum 2026* (p. 1475). <https://doi.org/10.2514/6.2026-1475>
- **Acharya K**, Sharif I, Lad M, Sun L, Song H, "Integrating neurosymbolic AI in advanced air mobility: a comprehensive survey", In *Proceedings of the Thirty-Fourth International Joint Conference on Artificial Intelligence (IJCAI '25)*. Article 1151, 10362–10370. <https://doi.org/10.24963/ijcai.2025/1151>
- **Acharya K**, Lad M, Sun L and Song H, "Neurosymbolic Approach for Travel Demand Prediction: Integrating Decision Tree Rules into Neural Networks", 2025 International Wireless Communications and Mobile Computing (IWCMC), Abu Dhabi, United Arab Emirates, 2025, pp. 600-605, doi: 10.1109/IWCMC65282.2025.11059465.
- **Acharya K**, Lad M, Sun L and Song H, "A Data-Driven Approach to Enhancing Gravity Models for Trip Demand Prediction", 2025 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2025, pp. 815-820, doi: 10.1109/CAI64502.2025.00145.

- **Acharya K**, Lad M, Song H, Sun L, "Regional air mobility flight demand modeling in tennessee state", In AIAA SCITECH 2025 Forum 2025 (p. 2783). <https://doi.org/10.2514/6.2025-2783>
- Hakim S.B, Adil M, **Acharya K**, Song H, "Decoding Android Malware with a Fraction of Features: An Attention-Enhanced MLP-SVM Approach", In: Song, H.H., Di Pietro, R., Alrabaei, S., Tubishat, M., Alkafairy, M., Alfandi, O. (eds) Network and System Security. NSS 2024. Lecture Notes in Computer Science, vol 15564. Springer, Singapore. https://doi.org/10.1007/978-981-96-3531-3_10
- Lad M, **Acharya K**, Sun L and Song H, "Enhancing Forecasting for Advanced Air Mobility", 2024 IEEE International Conference on Big Data (BigData), Washington, DC, USA, 2024, pp. 2908-2913, doi: 10.1109/BigData62323.2024.10825414.
- **Acharya K**, Lad M, Sun L and Song H, "Demand Modeling for Advanced Air Mobility", 2024 IEEE International Conference on Big Data (BigData), Washington, DC, USA, 2024, pp. 2855-2863, doi: 10.1109/BigData62323.2024.10825121.
- **Acharya K**, Velasquez A, Liu Y, Liu D, Sun L and Song H, "Improving Air Mobility for Pre-Disaster Planning with Neural Network Accelerated Genetic Algorithm", 2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC), Edmonton, AB, Canada, 2024, pp. 1290-1295, doi: 10.1109/ITSC58415.2024.10920105.
- Zhu L, Lan Q, Velasquez A, Song H, **Acharya K**, Tian Q, Niu S, "SKGHOI: Spatial-Semantic Knowledge Graph for Human-Object Interaction Detection," 2023 IEEE International Conference on Data Mining Workshops (ICDMW), Shanghai, China, 2023, pp. 1186-1193, doi: 10.1109/ICDMW60847.2023.00155.

TEACHING EXPERIENCE & MENTORSHIP

- Over 10 years of university-level teaching experience in computer science, engineering, artificial intelligence, networking, and software development courses.
- Taught undergraduate and graduate-level courses including Programming (C/C++/Python), Data Mining and Artificial Intelligence, Computer Architecture, Digital Logic Design, Data Communication, and Advanced Internetworking.
- Supervised undergraduate capstone and research projects involving AI, software systems, and data-driven applications.
- Designed course materials, assessments, and project-based learning activities emphasizing practical and research-oriented learning.

PROFESSIONAL SERVICE

Professional Leadership

- IEEE Senior Member and Vice-Chair Elect, IEEE Baltimore Section (2026–Present)
- IEEE Day Ambassador, IEEE Region 2 (2025)
- Region 2 Lead, IEEEExtreme 18.0, 19.0, and 20.0

Conference Leadership

- Registration Chair, ACM TRUST 2027: ACM Conference on Trustworthy and Responsible AI and Computing Systems, Washington, DC, USA
- Co-Chair, IEEE Baltimore Technical Colloquium & Professional Development Conference, 2026
- Conference Secretary, IEEE Baltimore Technical Colloquium & Professional Development Conference, 2025
- Technical Program Committee Member, The Twenty-First International Multi-Conference on Computing in the Global Information Technology (ICCGI 2026)

Editorial Appointments

- Newsletter Editor, ACM EIGTRUST (Emerging Interest Group on Trustworthy AI and Autonomous Systems), Association for Computing Machinery (ACM), 2026–Present

Journal Reviewer

- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Transactions on Artificial Intelligence
- IEEE Transactions on Intelligent Transportation Systems
- IEEE Internet of Things Journal
- Journal of Big Data
- IEEE Transactions on Consumer Electronics
- IEEE SMC Magazine
- Arabian Journal for Science and Engineering
- KSII Transactions on Internet and Information Systems

HONORS & AWARDS

- Conference Travel Grant (\$1000), Department of Information Systems, UMBC, 2026
- Professional Development Grant (\$750), Graduate Student Association, UMBC, 2026
- Outstanding Reviewer (2024–2025), IEEE Transactions on Neural Networks and Learning Systems, IEEE Computational Intelligence Society, Dec 2025
- Doctoral Research Poster Award, UMBC, May 2025
- IEEE CIS Publication Spotlight Recognition, IEEE Computational Intelligence Society, Nov 2024
- Elected to Phi Kappa Phi Honor Society

INVITED TALKS

- “Neurosymbolic AI for Advanced Air Mobility: Foundations, Technical Advances, and an Approach to Travel Demand Modeling,” IS 792 – Advanced Topics in Information Systems, UMBC, Nov 20, 2025.